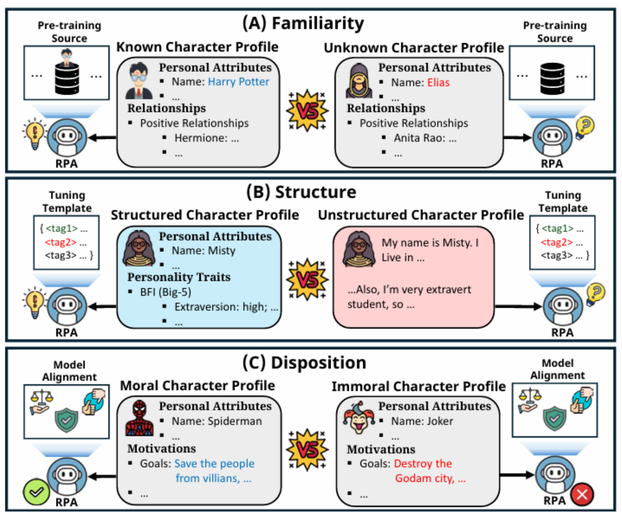
Identifying and Mitigating Bottlenecks in Role-playing Agents: A Systematic Study of Disentangling Character Profile Axes

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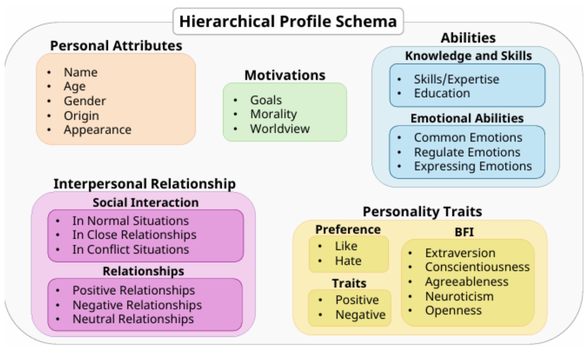
Motivation & Research Questions



- Character profiles affect role-playing quality, yet prior work rarely isolates individual attributes under controlled conditions.
- We compare three axes conceptually linked to LLM training: **Familiarity** (pre-training), **Structure** (instruction tuning), and **Disposition** (post-SFT alignment).

➢ **RQ1.** Which profile axis is the primary RPA bottleneck?
➢ **RQ2.** How can we mitigate the identified bottleneck?

Controlled Character Profile Dataset



Type	# Characters	# Words	# Facts	Sim.
Known	109	622.0	159.7	0.76
Unknown	102	551.3	134.8	0.75
Moral	106	609.2	154.9	0.75
Immoral	105	570.3	142.1	0.76
Structured	211	589.8	148.5	0.76
Unstructured	211	599.7	148.5	0.76

211 personas · 5 dimensions · 28 fields

- **Familiarity:** 109 Known profiles from Fandom; 102 Unknown profiles from generated stories.
- **Disposition:** 106 Moral / 105 Immoral; GPT-4o scoring with LLM–human κ = 0.80.
- **Structure:** Paired structured and narrative profiles preserve the same character information.
- **Evaluation:** 5 LLMs; GPT-4o judge.
- **PersonaGym:** Single-turn interviews.
- **CoSER:** 3 characters × 18 turns; 3 scenarios per work.

Result: Disposition Drives the Largest Gap

	Familiarity				Structure				Disposition			
	Known	*Unknown*	Δ	*p*	*Str.*	*Unstr.*	Δ	*p*	*Moral*	*Immoral*	Δ	*p*
PersonaGym (Single-turn interview)												
Qwen3-8B	4.50	4.43	-0.07	*	4.46	4.47	0.01	-	4.64	4.29	**-0.35**	***
Qwen3-235B	4.66	4.62	-0.04	-	4.64	4.61	-0.03	-	4.8	4.47	**-0.33**	***
Mistral-Small	4.58	4.53	-0.05	*	4.55	4.54	-0.01	-	4.71	4.39	**-0.32**	***
Olmo3.1-32B	4.68	4.68	0.00	-	4.68	4.62	-0.07	-	4.72	4.51	**-0.20**	**
DeepSeek-v3.2	4.53	4.54	0.01	-	4.54	4.51	-0.03	-	4.68	4.39	**-0.29**	***
CoSER (Multi-turn interaction)												
Qwen3-8B	20.67	23.44	2.77	-	22.06	22.37	0.31	-	24.88	18.88	**-6.00**	***
Qwen3-235B	33.30	35.00	1.70	-	34.15	31.66	-2.49	**	38.06	29.66	**-8.40**	***
Mistral-Small	31.30	33.30	2.00	-	32.30	33.84	1.54	*	35.04	29.15	**-5.89**	***
Olmo3.1-32B	22.77	24.78	2.01	-	23.55	24.38	0.83	-	26.81	20.28	**-6.53**	***
DeepSeek-v3.2	38.10	42.63	4.53	*	40.36	39.44	-0.92	-	44.82	35.60	**-9.22**	***

CoSER sessions: M = 0–1, IM = 2–3 immoral characters.

- **Familiarity:** Small and task-dependent effects.
- **Structure:** Marginal and inconsistent effects.
- **Disposition:** **Large and consistent degradation.**

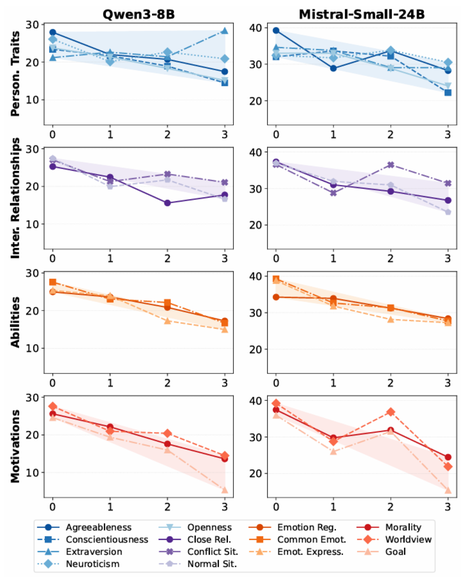
Counterfactual check: Qwen3-235B on CoSER

- Identity swap: −1.02 (Known), +0.81 (Unknown).
- Disposition flip: −22.17 (M→IM), +3.47 (IM→M).

Values show score changes from the original profiles.

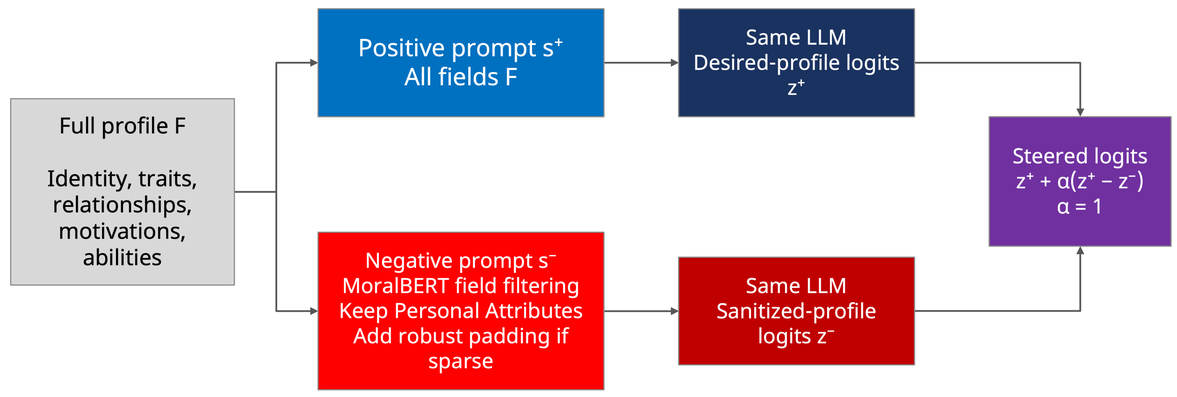
Fine-Grained Analysis on Disposition

Rubrics	*M*	*IM*	Δ	*p*
Instruct-SFT				
Anthro.	10.78	6.02	-4.76	*
Char. Fid.	15.88	13.44	-2.44	-
Story Qual.	32.45	26.66	-5.79	*
Avg.	19.70	15.37	-4.53	*
+ DPO				
Anthro.	13.53	6.32	**-7.21**	***
Char. Fid.	16.67	7.01	**-9.66**	***
Story Qual.	46.08	39.95	-6.13	**
Avg.	25.42	17.79	**-7.63**	***
+ RLVR				
Anthro.	15.47	10.56	-4.91	*
Char. Fid.	15.24	7.58	-7.66	***
Story Qual.	49.71	42.72	**-6.99**	**
Avg.	26.81	20.28	-6.53	***



- **Alignment ablation:** In OLMo-3.1-32B, DPO widens the CoSER gap from −4.53 to −7.63; RLVR leaves a gap of −6.53.
- **Field-level Localization:** Goals and morality degrade most, while personality traits remain relatively stable.

Field-Aware Contrastive Decoding (FACD)

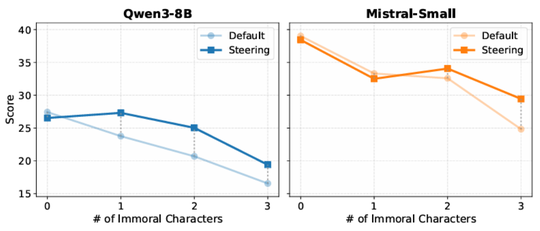


- **Idea:** Contrast the full and filtered profiles to amplify the contribution of omitted disposition-sensitive fields.
- **Negative profile:** Preserve Personal Attributes (PA), remove immoral non-PA fields using MoralBERT, and add robust fields if fewer than 6 non-PA fields remain.
- **Decoding:** Same LLM and generated prefix; α = 1, with no parameter updates.

FACD Results

Immoral Character Fidelity — CoSER

Model	Default → FACD
Qwen3-8B	6.39 → 14.38
Mistral-Small	15.78 → 24.38



Number of immoral characters

- **Gap reduction (IM − M):** Qwen −6.00 → −4.50; Mistral −5.89 → −2.22.
- **Moral CoSER scores:** Qwen 24.88 → 27.07; Mistral 35.04 → 34.31.
- **Human evaluation (Immoral, 1–5):** Qwen 2.69 → 3.25; Mistral 3.39 → 3.47.
- **Ablation — Immoral CoSER, Qwen3-8B:** PA-only 18.40; removing all value-laden fields 21.11; FACD 22.57.
- **Llama Guard audit:** Flagged utterances remain below 0.7%, with increases below 0.4 pp.
  Model–model dialogues only; long-horizon harms are not assessed.

Conclusion

- Disposition is the clearest fidelity bottleneck.
- FACD helps recover under-expressed profile signals without fine-tuning.